One of the most popular scheme of this type is the technique of dissipative particle dynamics (DPD) which was introduced in 1992 by Hoogerbrugge and Koelman [181]. DPD groups several fluid molecules and treats these as a single particle (which may be thought of as a lump of fluid). The particles of a DPD system interact through a short-ranged "soft" pairwise (distance-based) potential energy function. The use of such a soft potential increases the usable timestep, thus the benefit of using DPD as opposed to MD is felt both in reduction of system dimension and increase in the timestep and thus the accessible time interval in simulation [156]. DPD can be viewed as a Langevin dynamics in which configuration dependence is present in both dissipative and stochastic perturbations. We will consider numerical methods for DPD in this section; for discussion of coarse-graining applications of DPD, see for example [39, 88, 330].

The equations of DPD are a little more complicated than standard MD. It is therefore important to design accurate and efficient methods tailored to their special form. Most numerical methods for DPD simulation exhibit substantial statistical bias. Some methods have been developed to reduce the per timestep efficiency but at a price in terms of the conservation properties. The only way that the errors are tamed in some ad hoc schemes is by reducing the timestep size excessively (well below the stability threshold), but this may destroy the practical value of those methods.

In this section we present numerical methods based on splitting for DPD and for another stochastic momentum-conserving method, the pairwise Nosé-Hoover-Langevin (PNHL) method.

8.9.1 Formulation of DPD

We describe here the formulation of dissipative particle dynamics due to Español and Warren [122].

Let the N particles be described in the usual way by positions $\boldsymbol{q}_i$ and momenta $\boldsymbol{p}_i$. We assume their masses to be m_i for $i = 1, \ldots, \mathrm{N}$. The DPD equations of motion are of the form

$$\frac{\mathrm{d}\boldsymbol{q}_i}{\mathrm{d}t} = \frac{\boldsymbol{p}_i}{m_i}, \quad \frac{\mathrm{d}\boldsymbol{p}_i}{\mathrm{d}t} = \boldsymbol{F}_i, \tag{8.80}$$

where $\boldsymbol{F}_i$ is the total interparticle force acting on particle i due to the presence of the other particles. The force is composed of three pairwise contributions

$$\boldsymbol{F}_i = \sum_{j \neq i} (\boldsymbol{F}_{ij}^{\mathrm{C}} + \boldsymbol{F}_{ij}^{\mathrm{D}} + \boldsymbol{F}_{ij}^{\mathrm{R}}), \tag{8.81}$$

where $\boldsymbol{F}_{ij}^{\mathrm{C}}$, $\boldsymbol{F}_{ij}^{\mathrm{D}}$ and $\boldsymbol{F}_{ij}^{\mathrm{R}}$ represent conservative, dissipative and random forces, acting on particle i due to the j-th particle, respectively.

Many choices are possible for the conservative force field [219, 244, 342]. In case the DPD equations are used simply to control the momentum (and not as a coarse-graining device), any empirical molecular force field may be used. When DPD is used for coarse-graining, it is common to adopt a simplified pair potential, namely

$$\boldsymbol{F}_{ij}^{\mathrm{C}} = \begin{cases} \alpha\left(1 - \dfrac{r_{ij}}{r_c}\right)\hat{\boldsymbol{q}}_{ij}, & r_{ij} < r_c; \\ 0, & r_{ij} \geq r_c, \end{cases} \tag{8.82}$$

where r_{ij} is the interparticle distance, $\hat{\boldsymbol{q}}_{ij} = r_{ij}^{-1}(\boldsymbol{q}_i - \boldsymbol{q}_j)$, parameters α define the strength of the repulsion of particles i and j, and r_c is the cutoff radius. The use of a cutoff is important for efficiency reasons—in a large system, only the particles interacting at close range will generate nonzero forces of interaction so the force calculation may be substantially reduced. Both α and r_c could be allowed to be different for different particle pairs, although they are most often taken to be homogeneous. The pair potential corresponding to the force is

$$\varphi_{\mathrm{DPD}}(r) = \begin{cases} \dfrac{\alpha r_c}{2}\left(1 - \dfrac{r}{r_c}\right)^2, & r < r_c; \\ 0, & r \geq r_c. \end{cases} \tag{8.83}$$

so that the conservative potential energy is:

$$U_{\mathrm{DPD}} = \sum_{i<j} \varphi_{\mathrm{DPD}}(r_{ij}), \qquad r_{ij} = \|\boldsymbol{q}_i - \boldsymbol{q}_j\|. \tag{8.84}$$

The choice of dissipative and random forces is made so as to maintain a prescribed temperature. The typical formulas used for this purpose are

$$\boldsymbol{F}_{ij}^{\mathrm{D}} = -\gamma\omega^{\mathrm{D}}(r_{ij})(\hat{\boldsymbol{q}}_{ij} \cdot \boldsymbol{v}_{ij})\hat{\boldsymbol{q}}_{ij}, \tag{8.85}$$

$$\boldsymbol{F}_{ij}^{\mathrm{R}} = \sigma\omega^{\mathrm{R}}(r_{ij})\theta_{ij}\hat{\boldsymbol{q}}_{ij}, \tag{8.86}$$

where γ and σ parameterize the forces (again, they could be taken to be different for different pairs), and ω^{D} and ω^{R} are configuration-dependent weight functions that typically incorporate a cut-off similar to that used in the conservative force, e.g. one may take

$$\omega^{\mathrm{R}}(r) = \begin{cases} 1 - \dfrac{r}{r_c}, & r < r_c; \\ 0, & r \geq r_c, \end{cases} \tag{8.87}$$

where r_c is the same cutoff used in the conservative force. The relative velocities are denoted by $\boldsymbol{v}_{ij} = \boldsymbol{p}_i/m_i - \boldsymbol{p}_j/m_j$, and θ_{ij} is a symmetric ($\theta_{ij} = \theta_{ji}$) Gaussian